

Manufacturing Station Operation Manual

System Overview:

This is a control panel interface that controls multiple processing stations to process raw materials and produce products.

System Components and Workflow Introduction (Figure 1):

- **Raw Material:**

The material entered into the processing stations, which is then processed into a product.

Note: Raw materials are represented as integers.

- **Product:**

A product generated after raw material passes through the processing stations.

Note: Like raw materials, products are represented as integers.

- **Processing Station:**

A station that processes raw materials into products.

Note: Different processing stations contain different processing logic, as described below.

The processing stations include the following types:

- ♦ **Station A:**

- **Adds 1** to the raw material.

- ♦ **Station B:**

- **Subtracts 1** from the raw material.

- ♦ **Station C:**

Performs different processing logics depending on the received raw material:

- If the raw material is **odd**, it is sent to the next station.
- If the raw material is **even**, the next station is skipped, and it is sent to the station after that.

- ♦ **Station D:**

Performs different processing logics depending on the received raw material:

- If the raw material is **odd**, it **adds 1**.
- If the raw material is **even**, it **adds 2**.

- **Control Panel**

The control center of the entire production workflow.

1. Enter raw materials.
2. Set the processing flow of the stations.
3. Output products.

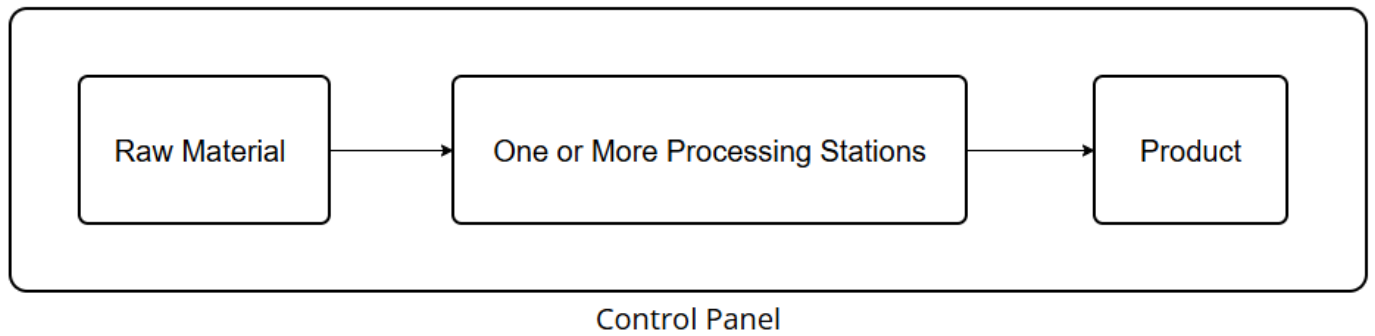


Figure 1. System Workflow Diagram

Control Panel Functions (Figure 2):

1. Set processing flow:

Set the processing flow of the stations. e.g., A->B, C->B->A, A->B->C->B->A, etc.

Note: No spaces are allowed between the arrows and station names.

2. Input raw material:

Enter a raw material value. e.g., 1, 20, 61, etc.

3. Show current processing flow:

Show the currently configured processing flow.

4. Show all products:

Show all products that have been produced.

5. Show station processing counts:

Show how many times each processing station has processed materials.

6. Reset application state:

Clear all previous production records, including:

- ♦ Produced products
- ♦ Processing count of each station
- ♦ Current processing flow

7. Exit and save:

Save the processing results and close the application. When the application is opened again, the saved processing results can be restored.

```
Manufacturing Station Console
1. Set processing flow
2. Input raw material
3. Show current processing flow
4. Show all products
5. Show station processing counts
6. Reset application state
7. Exit and save
Please choose:
```

Figure 2. Control Panel Interface

System Operation Example:

1. Open the Console by executing **ADEQuiz.exe**. After startup, the system directly enters the Control Panel and displays the available functions.
2. **Select 1. Set processing flow.** Then set the processing flow to: A->A
3. **Select 2. Input raw material.** Then enter: Raw material = 1
4. The resulting product is: 3
5. **Select 1. Set processing flow.** Then set the processing flow to: A->B->C->B->A
6. **Select 2. Input raw material.** Then enter: Raw material = 2
7. The resulting product is: 3
8. **Select 4. Show all products.** The system displays all produced products and the total number of products.

Example output:

Products:

1. 3
2. 3

Total: 2 products

9. Select **5. Show station processing counts.** The system displays how many times each station has processed materials. **Note: If a station is skipped by Station C, it will not be counted.**

Example output:

Station A processed 4 times
Station B processed 1 time
Station C processed 1 time

10. Select **7. Exit and save.** The application exits and saves the processing results.

Note: If a product is successfully produced, the system automatically saves the processing results.

Additional Function Examples

- ♦ **Select 3. Show current processing flow.** The system displays the currently configured processing flow. Continuing from the previous example, the final processing flow is: A->B->C->B->A. The system output is:

Current processing flow: A->B->C->B->A

- ♦ **Select 6. Reset application state.** This clears all previous production records. Since this is a relatively risky operation, the system asks the operator for confirmation before proceeding.